

Assignment 2: Transfer Functions and Modeling

The textbook exercises, designated with the word “Nise”, are from the “Problems” section at the end of Chapter 2 in *Control System Engineering* by Norman Nise, 7th edition.

Nise reading:

- Chapter 2, Sections 2.5 and 2.6
- Appendix B, ch2p9–ch2p12

Inverse Laplace Transform of Functions

1. Given the following Laplace transforms, find the corresponding time functions. Use the MATLAB `residue` function for the partial fraction expansion. You do not need to turn in your MATLAB scripts.

$$\text{a) } F(s) = \frac{s+5}{(s+2)(s^2+36)}$$

$$\text{b) } G(s) = \frac{2}{s(s^2+s+16.25)}$$

$$\text{c) } H(s) = \frac{1}{(s^2+9)(s^2+4s+1)}$$

Transfer Functions

2. Nise 2.8
3. Nise 2.9

Transfer Function Representation in MATLAB

4. Consider each of the following transfer functions:

$$\text{a) } G(s) = \frac{3}{s^3+9s^2+20s}$$

$$\text{b) } H(s) = \frac{10(s+2)}{s^2(s+4)(s+5)}$$

For each expression, illustrate (submit a snippet of MATLAB script) how you would generate the transfer function object in MATLAB using the two methods:

- using the ratio of factors (the `zpk()` function)
- using the ratio of polynomials (the `tf()` function)

5. Nise 2.13

Note: the `roots` function can be used to find the roots of a polynomial. For example, to find the roots of $s^2+5s+30$ we could do the following...

```
>> roots([1 5 30])
ans =
```

$$\begin{aligned} & -2.5000 + 4.8734i \\ & -2.5000 - 4.8734i \end{aligned}$$

USV Model

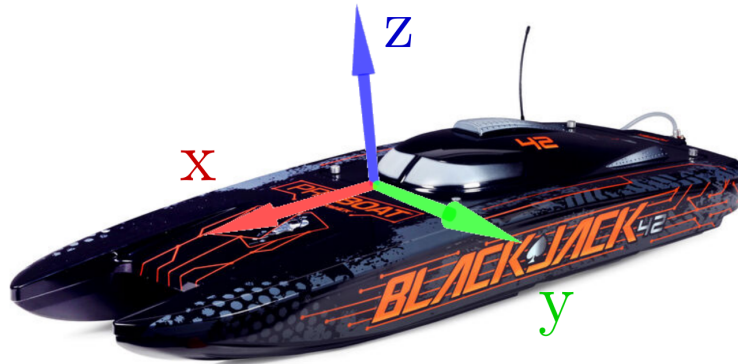


Figure 1: Small unmanned surface vessel (USV) with its body-fixed coordinate frame: x forward, y port, z up.

Consider the small unmanned surface vessel (USV) in Figure 1. The vessel is approximately 1.1 m long with a mass of 5 kg. It has a single propeller on the centerline, which produces a thrust force $F(t)$ [N] along the body x -axis, and a single stern rudder, deflected by an angle $\delta(t)$ [rad], which produces a yaw moment $N(t)$ [N·m] about the body z -axis.

The vessel has six degrees of freedom: three translations (surge, sway and heave) and three rotations (roll, pitch and yaw). Because the propeller is on the centerline, we treat the two inputs as decoupled: thrust controls forward speed, and the rudder controls heading. We model only two scalar motions:

- Surge velocity $v_x(t)$ [m/s], forward velocity along the body x -axis.
 - Yaw rate $r(t) = \dot{\psi}(t)$ [rad/s], where $\psi(t)$ [rad] is the heading angle.
6. Consider the surge direction only. Find the equation of motion in the surge ($+x$) direction with the following assumptions:
- The total inertia (rigid body plus added mass) of the vessel in the surge direction is $m = 5.0$ [kg].
 - The total drag on the vessel is linearly related to surge velocity, with a coefficient of $b_x = 8.0$ [N·s/m].
 - The only other force in the surge direction is the propeller thrust $F(t)$ [N].
7. Consider the scenario where the vessel starts at rest with no input. At time zero the propeller thrust is set to a constant $F(t) = 12.0$ [N].
- Using Laplace transforms, solve the equation of motion to find the surge velocity v_x as a function of time.
 - Using the expression obtained, plot the time response of the surge velocity v_x versus time t . Include labels and units on the axes.

- What is the steady-state speed of the vessel? Indicate it on your plot.
 - How long does it take for the vessel to reach 95% of the steady-state speed? (An estimate from the graph is sufficient.)
8. Find the transfer function for the surge direction where the input is the thrust F and the output is the surge velocity v_x .
 9. Next, consider the yaw direction only, and assume the vessel is moving at a constant forward speed. Find the equation of motion in the yaw direction with the following assumptions/approximations:
 - The total inertia (rigid body plus added inertia) of the vessel about the yaw axis is $I = 0.8$ [kg·m²].
 - The drag moment on the vessel is linearly related to yaw rate, with a coefficient of $b_r = 1.2$ [N·m·s/rad].
 - The only other moment about the yaw axis is produced by the rudder. For small deflection angles at a constant forward speed, the moment is proportional to the deflection angle. The rudder moment is therefore approximately proportional to the rudder deflection, $N(t) = N_\delta \delta(t)$, with rudder effectiveness $N_\delta = 2.0$ [N·m/rad]. The input to the yaw model is the rudder deflection $\delta(t)$ [rad].
 10. Find the transfer function for the yaw direction where the input is the rudder deflection $\delta(t)$ and the output is the yaw rate $r(t)$.
 11. Find the transfer function for the yaw direction where the input is the rudder deflection $\delta(t)$ and the output is the heading angle $\psi(t)$.
 12. Consider the scenario where the vessel starts with zero yaw rate and zero heading. At time zero the rudder is deflected to a constant angle $\delta(t) = 0.3$ [rad] and held there.
 - Use the final value theorem to find the steady-state yaw rate.